

Abhishek Sahoo, M.S.

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SUMMARY

- Biomedical scientist with expertise in epigenomics, radiation biology, and stem cell research, with a publication record spanning 7 peer-reviewed articles. Independently designs and executes complex studies integrating molecular, computational, and in vivo approaches across cGMP-compliant and clinical-translational settings. Invited speaker at the Gordon Research Seminar and recognized with multiple national awards for research excellence.

EXPERIENCE

Graduate Research Assistant

UT-ORNL Bredeesen Center

Aug 2023 – Present

Knoxville, TN, USA

- Lead independent research on the molecular consequences of low-dose radiation using epigenetics in lung fibroblasts and epithelial cells, in collaboration with Oak Ridge National Laboratory.
- Investigated changes in histone marks H3K27me3, H3K9me3, and H3K27ac following low-dose radiation exposure using CUT&RUN.
- Handled radioisotopes Cs137 and Ac225 for gamma and alpha exposure studies in lung fibroblasts and epithelial cells.
- Developed computational pipelines to process RNA-seq, CUT&RUN, and Hi-C data on a high-performance computing cluster (ISSAC-NG) using Python, R, DESeq2, deepTools, and Bowtie2.
- Led synthesis of drug- and radioisotope-encapsulated PLGA nanoparticles to assess anti-cancerous efficacy and targeted alpha therapy.
- Visualized fibrosis, EMT, and DNA damage through confocal immunofluorescence microscopy in untreated and radiation-exposed cells.
- Maintained lab data and adhered to OSHA, CDC, and NIH BSL-2 guidelines for handling human cell lines.

Junior Research Fellow

L V Prasad Eye Institute

Jan 2021 – Jul 2023

Hyderabad, India

- Established culture and expansion of primary mesenchymal stem cells from the limbus of human cornea under sterile GMP conditions.
- Characterized cultured cells in xeno-free media for mesenchymal phenotype using immunofluorescence, qRT-PCR, flow cytometry, CFU assay, and wound healing assay.
- Managed in vivo studies using mice and rabbits to assess pre-clinical efficacy of cultured mesenchymal stem cells in corneal wound healing.
- Prepared and qualified the cell product and coordinated patient enrollment for clinical trial NCT04932629, evaluating cultured limbal stem cells in corneal stromal wounded patients.
- Authored two peer-reviewed research articles and co-authored five articles in international journals.
- Presented scientific findings as a poster at national and international conferences.

Master's Dissertation

Pondicherry University

Oct 2019 – Mar 2020

Pondicherry, India

- Independently evaluated hesperetin as a candidate therapeutic agent against neuroblastoma using an in silico approach and nano drug formulation.
- Synthesized silver nanoparticles encapsulated with hesperetin and characterized them using DLS.
- Conducted molecular docking using Autodock and PyMol to investigate binding affinity of hesperetin with principal pathway proteins in neuroblastoma.

Summer Intern

Institute of Life Sciences

Jun 2019 – Jul 2019

Bhubaneswar, India

- Performed cloning of the E-cadherin promoter in DH5alpha cells.
- Studied the effects of NAE and HER2 overexpression on E-cadherin promoter expression.
- Conducted western blotting to assess protein expression levels.

Graduate Teaching Assistant

University of Tennessee

Jan 2026 – May 2026

Knoxville, TN, USA

- Served as Teaching Assistant for a biology course in the Department of Biochemistry and Molecular Biology.
- Led in-class demonstrations on enzyme kinetics, the central dogma of molecular biology, and cell cycle and division.
- Facilitated weekly discussions on reading and interpreting scientific journal articles.
- Graded weekly homework, in-class group work, and contributed to exam grading.

SKILLS

Genomics and Epigenomics: CUT&RUN, RNA-seq, Hi-C (chromosome conformation capture), qRT-PCR, UCSC genome browser, STRING analysis

Computational: Python, R, DESeq2, deepTools, Bowtie2, HPC (ISSAC-NG), Autodock, PyMol, Prism, ImageJ

Molecular and Cell Biology: Mammalian cell culture, SDS-PAGE/Western blotting, ELISA, Immunostaining (IHC/ICC), Flow

cytometry, GMP cell culture

Imaging and In Vivo: Confocal microscopy, SEM microscopy, Mice and rabbit handling, PLGA nanoparticle synthesis, Theranostic

AWARDS & HONORS

- Department of Energy Travel Award (Apr 2026)
- SIT Travel Award, Radiation Research Society (Sep 2025)
- Outstanding Scholar Awardee, Radiation Research Society (Sep 2025)
- UTK GSS Travel Award (Oct 2025)
- Graduate Student Senate Award for Excellence in Graduate Research (May 2024)
- First Author Award, Bredesen Center (May 2024)

PUBLICATIONS

- Sahoo, A., Damala, M., Jaffet, J. et al. Expansion and Characterization of Human Limbus-derived Stromal/Mesenchymal Stem Cells (hLMSCs) in Xeno-free Medium for Therapeutic Applications. *Stem Cell Res Ther* 14, 89 (2023). <https://doi.org/10.1186/s13287-023-03299-3>
- Damala, M.; Sahoo, A. et al. Pre-Clinical Evaluation of Efficacy and Safety of Human Limbus-Derived Stromal/Mesenchymal Stem Cells with and without Alginate Encapsulation for Future Clinical Applications. *Cells* 2023, 12, 876. <https://doi.org/10.3390/cells12060876>
- Thacker, Minal; Sahoo, Abhishek et al. Benzalkonium chloride-induced dry eye disease animal models: Current understanding and potential for translational research. *Indian Journal of Ophthalmology* 71(4):1256-1262, April 2023. DOI: 10.4103/IJO.IJO_2791_22
- Prasad, Deeksha; Salman, Mohd; Reddy, A Aniruth; Jaffet, Jilu; Sahoo, Abhishek et al. A review of rabbit models of meibomian gland dysfunction and scope for translational research. *Indian Journal of Ophthalmology* 71(4):1227-1236, April 2023. DOI: 10.4103/IJO.IJO_2815_22
- Vijay Kumar Singh, Abhinav Reddy Kethiri, Tejaswini Pingali, Abhishek Sahoo et al. Development and validation of a reliable rabbit model of limbal stem cell deficiency by mechanical debridement using an ophthalmic burr. *Experimental Eye Research*, Volume 236, 2023, 109667. <https://doi.org/10.1016/j.exer.2023.109667>
- Sujithra Shankar, Minal Thacker, Abhishek Sahoo et al. Revisiting rabbit models for keratoconus: A long-term study on collagenase-induced disease progression. *Experimental Eye Research*, Volume 241, 2024, 109811. <https://doi.org/10.1016/j.exer.2024.109811>
- Abhinav Reddy Kethiri, Anahita Kate, Madhuri Amulya Koduri, Abhishek Sahoo et al. Immunological and histopathological changes in alkali burn-induced ocular surface pannus: Implication on success vs. failure of SLET. *Experimental Eye Research*, Volume 255, 2025, 110378. <https://doi.org/10.1016/j.exer.2025.110378>

CONFERENCE PRESENTATIONS

- Invited oral presentation, Gordon Research Seminar, Newry, Maine (July 2026)
- Poster presentation, Gordon Research Conference, Newry, Maine (July 2026)
- Talk, American Chemical Society Spring Meeting, Atlanta, Georgia (March 2026)
- Talk, Radiation Research Society Annual Meeting, Puerto Rico (September 2025)

LEADERSHIP AND SERVICE

- Vice President, Indian Student Association at UTK (Jul 2026 - Present)
- Chair, International Student Support Committee, UTK Graduate Student Senate (Jul 2025 - May 2026)
- President, Indian Student Association at UTK (Jul 2025 - Jun 2026)
- Secretary, Indian Student Association at UTK (Jul 2024 - Jun 2025)

EDUCATION

University of Tennessee / Oak Ridge National Laboratory, UT-ORNL Bredesen Center	Knoxville, TN, USA
<i>M.S. in Genome Science and Technology.</i>	<i>December 2026</i>
Pondicherry University	India
<i>M.S. in Biochemistry and Molecular Biology.</i>	<i>November 2020</i>
Utkal University	India
<i>B.S. in Zoology.</i>	<i>May 2018</i>